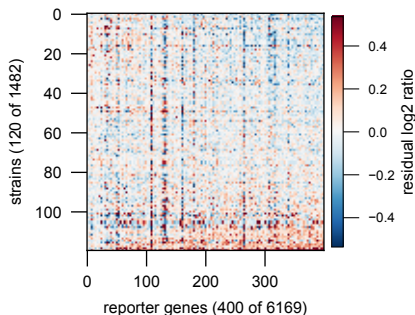
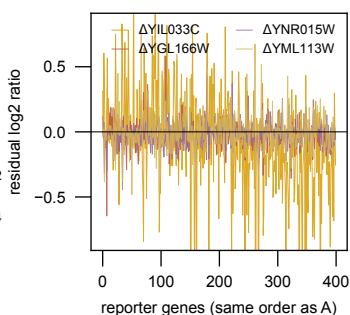


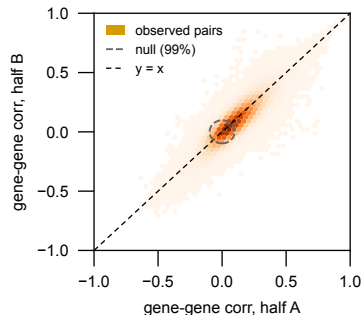
A the residual matrix R, log2 ratio
(1 deletion strain per row)



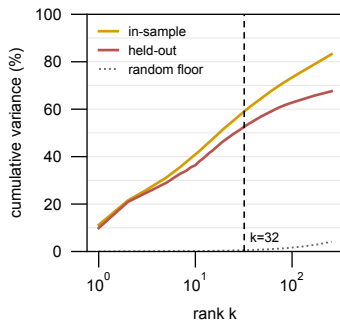
B four individual perturbations
(each row of A)



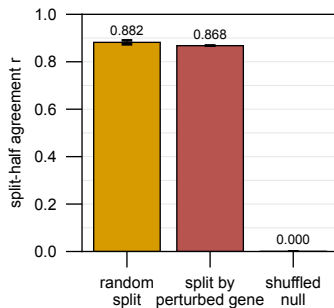
C floor check: structure exists
 $r=0.869$ vs null 0.0001



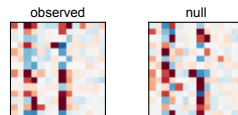
D which components GENERALIZE
(fit half A, score half B)



E THE RESULT: not conditional
on which gene was deleted



F what the null destroys



Each GENE COLUMN is permuted over strains INDEPENDENTLY. Every gene keeps its own values (same column histogram), but the pairing between columns is destroyed -- so any gene-gene correlation is removed while per-gene properties are held fixed.

This is a FLOOR, not the interesting test: it only asks whether cross-gene structure exists at all. Panel E is the real question.